Predicting *Mental Health Trends*

GROUP 11 - NeuroNexus

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Overview

- **Objective:** Analyze and predict mental health trends using Reddit discussions (r/anxiety, r/depression, r/adhd).
- **Data Collection:** Utilized PRAW API to fetch real-time Reddit posts, focusing on relevant subreddits.
- **Data Processing:** Preprocessed data by handling missing values, normalizing text, and removing duplicates.
- **Modeling Approach:** Employed machine learning models (Random Forest, XGBoost) for severity classification and clustering methods (LDA, K-Means) for topic modeling.
- **Diagnostic Tools:** Integrated PHQ-9, GAD-7, and DSM-5 assessments to enhance accuracy.
- **Expert Collaboration:** Consulted with mental health professionals to validate models and ensure reliability.
- **Impact:** Aims to enhance early detection, optimize mental health resource allocation, and support proactive interventions.

Overview of the Pipeline

- Fetch data from mental health subreddits.
- Clean and tokenize text.
- Perform lemmatization.
- Merge new data with existing datasets.

Data Collection (Reddit API)

- **Tools Used:** PRAW (Python Reddit API Wrapper)
- **Subreddits Scraped:** anxiety, depression, adhd
- **Key Features:**
  - Fetches the latest posts from subreddits.
  - Ensures no duplicate posts are collected.
  - Filters out posts older than 10 years.
  - Saves data in CSV format.

Data Cleaning & Tokenization

- Libraries Used: NLTK, spaCy
- Key Processing Steps:
- Convert text to lowercase.
- Tokenize words.
- Remove stopwords & punctuation.
- Quote each word for proper formatting.
- Replace missing authors with "Anonymous".

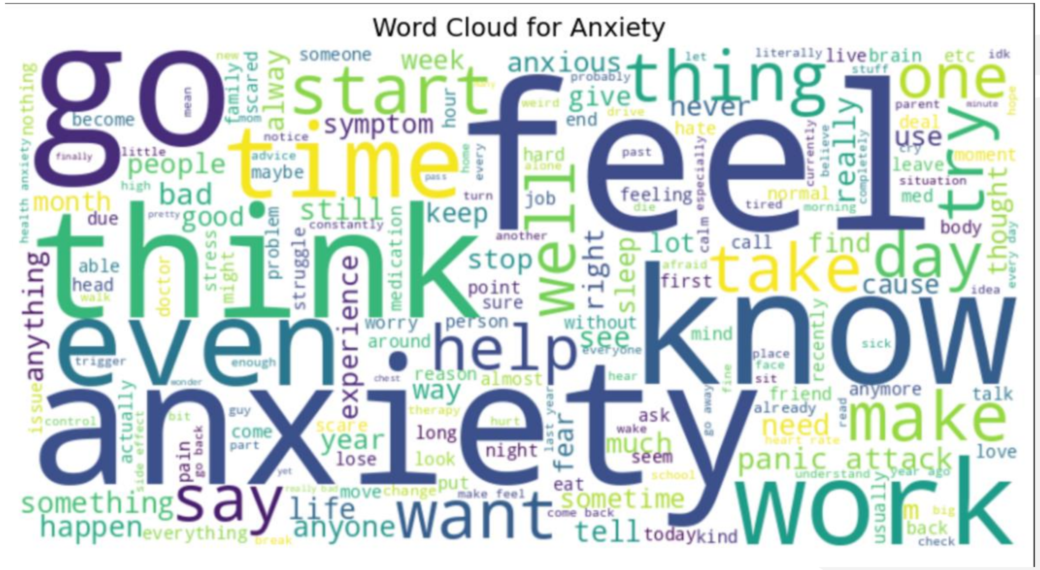
Lemmatization

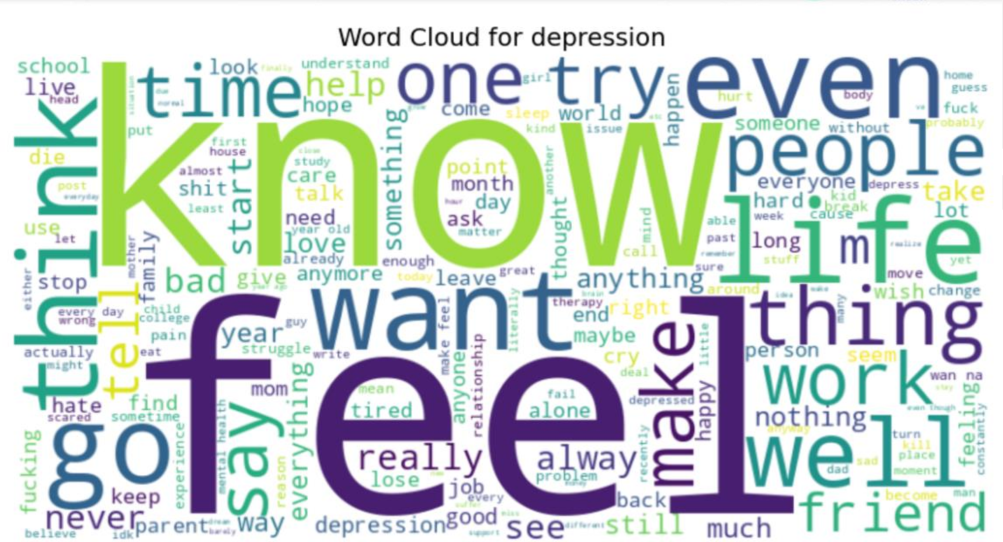
- Converts words to their base forms (e.g., "running" → "run").
- Helps improve text analysis.
- Tools Used: spaCy
- Code :
- doc = nlp(text)
- lemmatized_text = " ".join([f'"{token.lemma_}"' for token in doc])

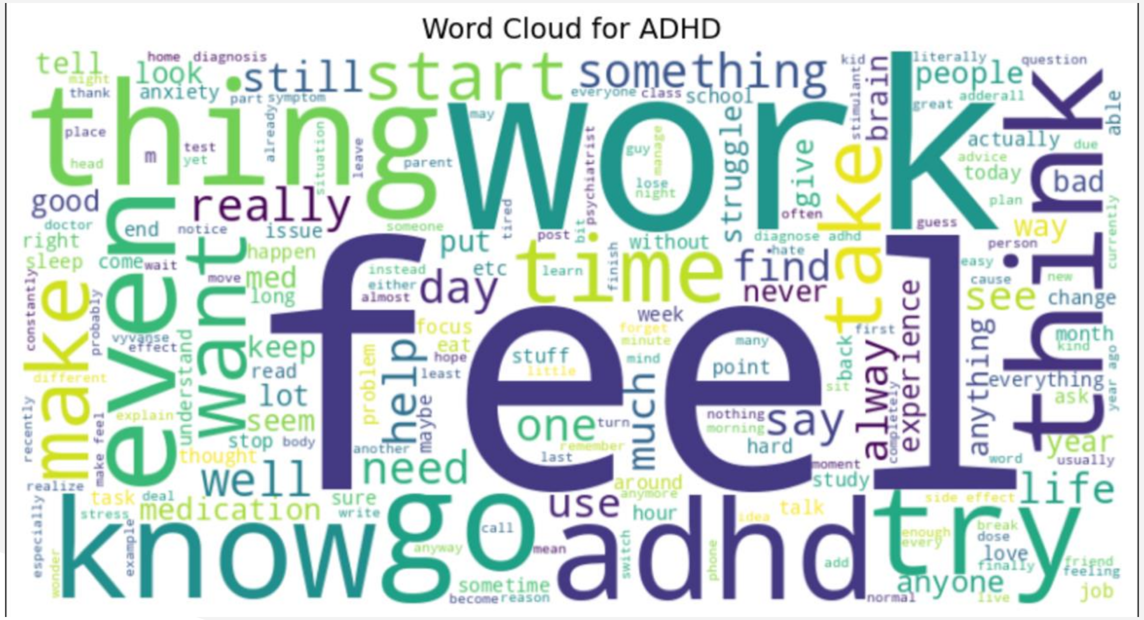
Exploratory Data Analysis (EDA)

1	Word Cloud for the separate diseases.
2	Word Cloud for Disease medications
3	Daily Post Frequency over Time
4	Top 10 active Users
5	Top 10 Users with Highest Engagement
6	Top 10 Ngrams for all the Diseases

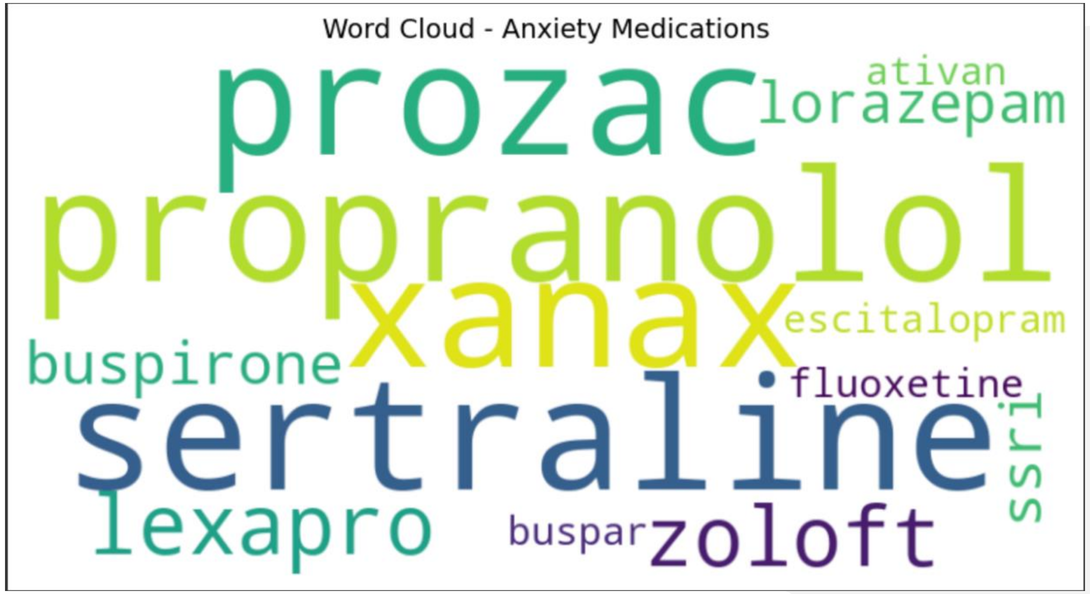
Word Cloud – All Diseases

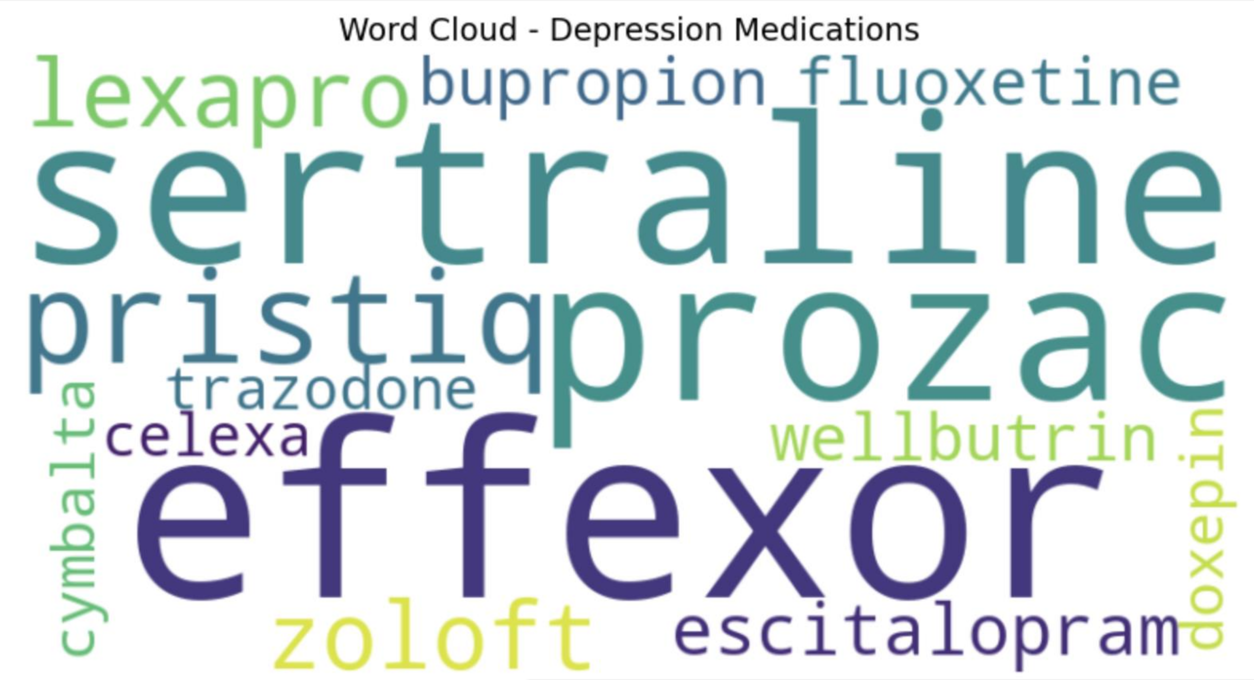


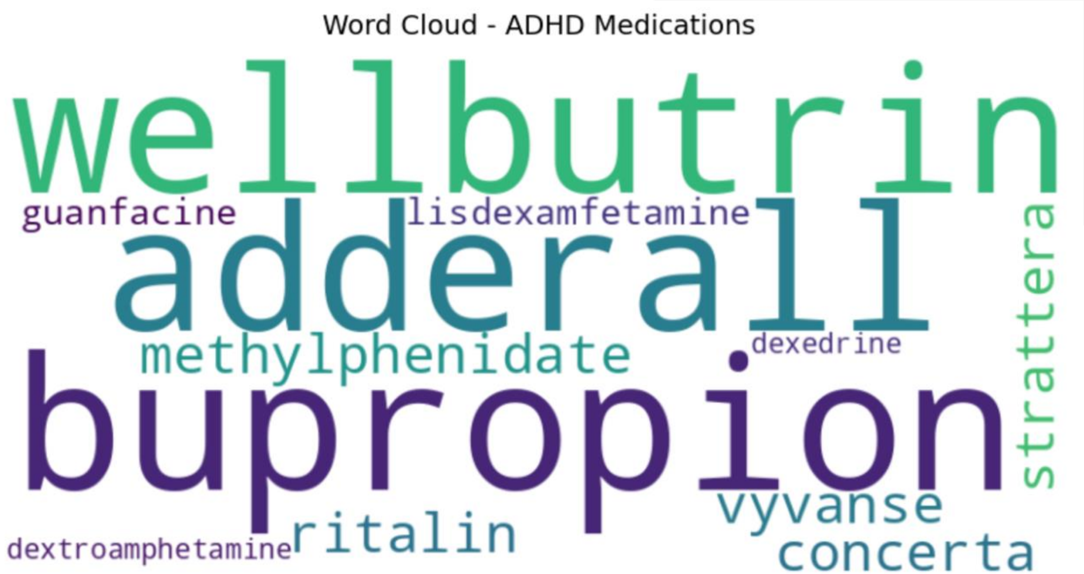




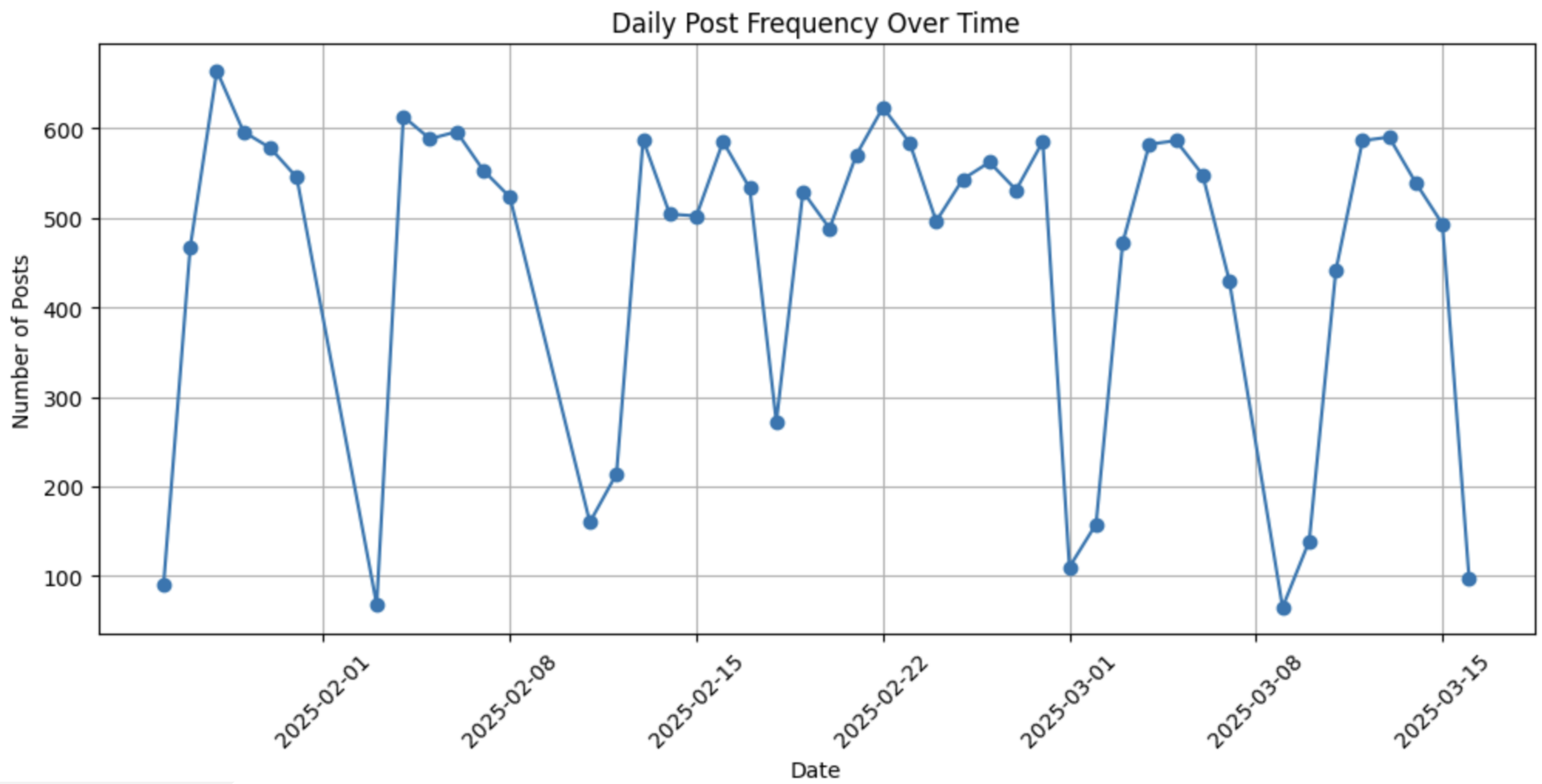
Word Cloud – Medications



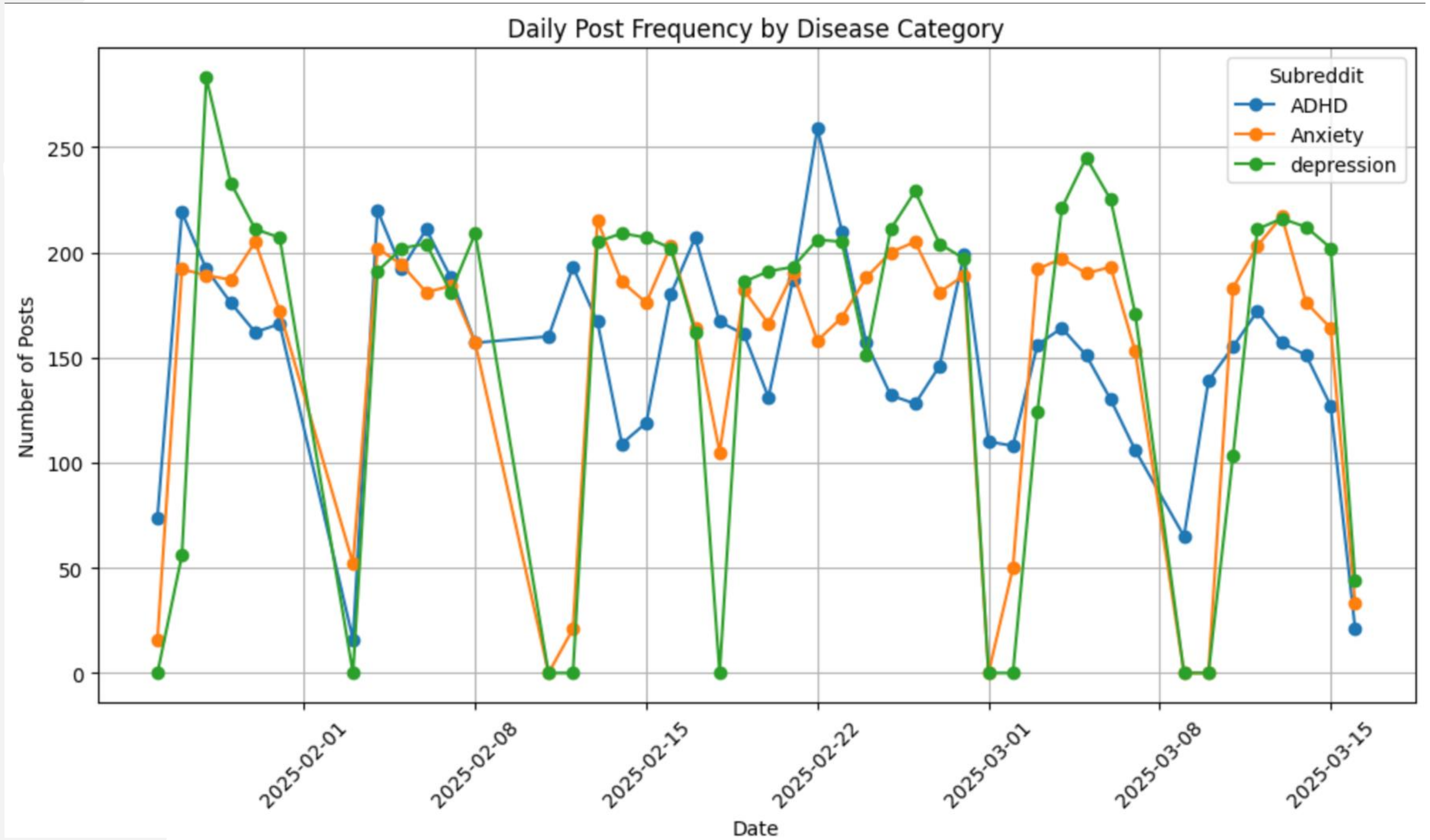




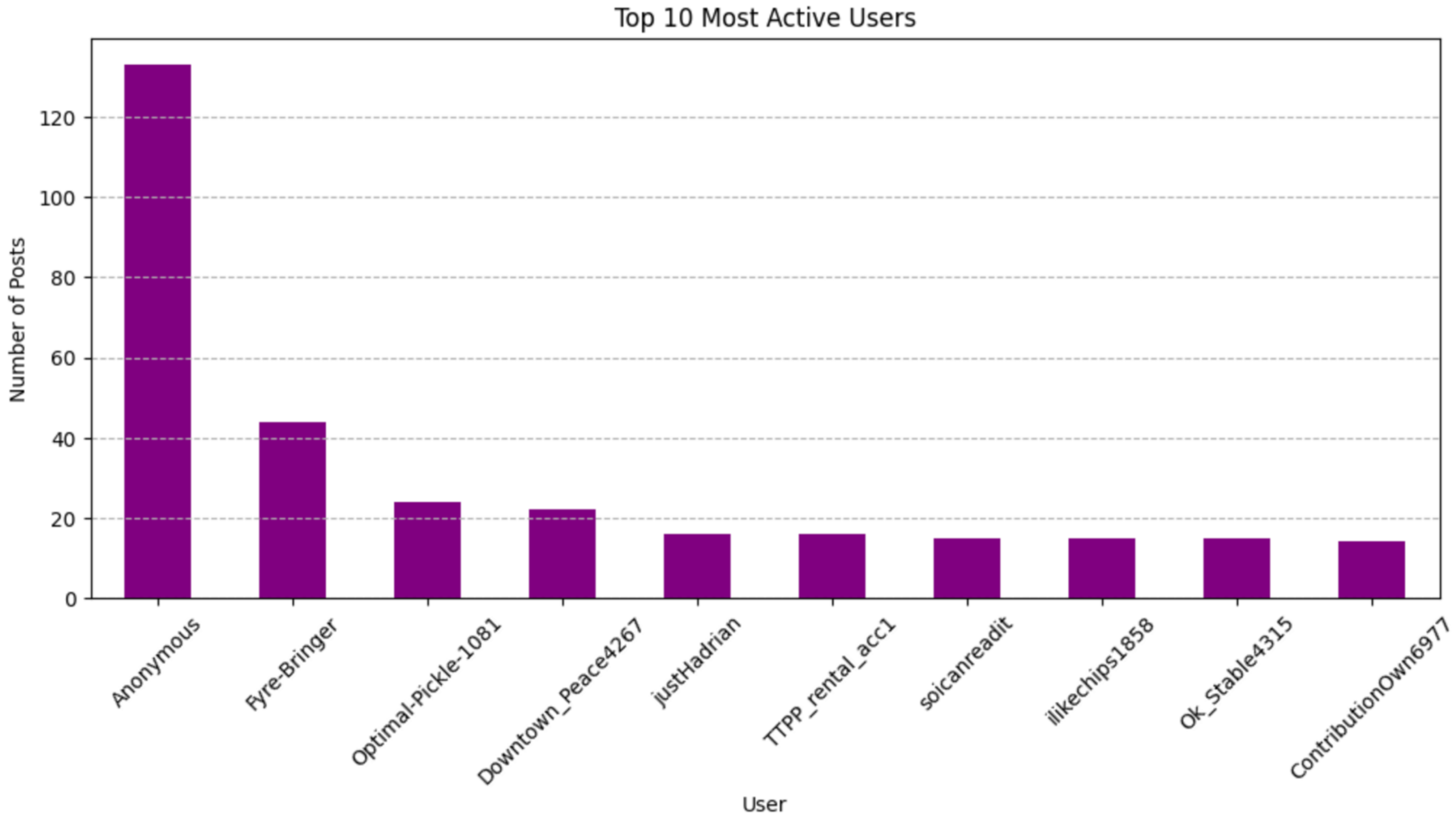
Daily Post Frequency Over Time



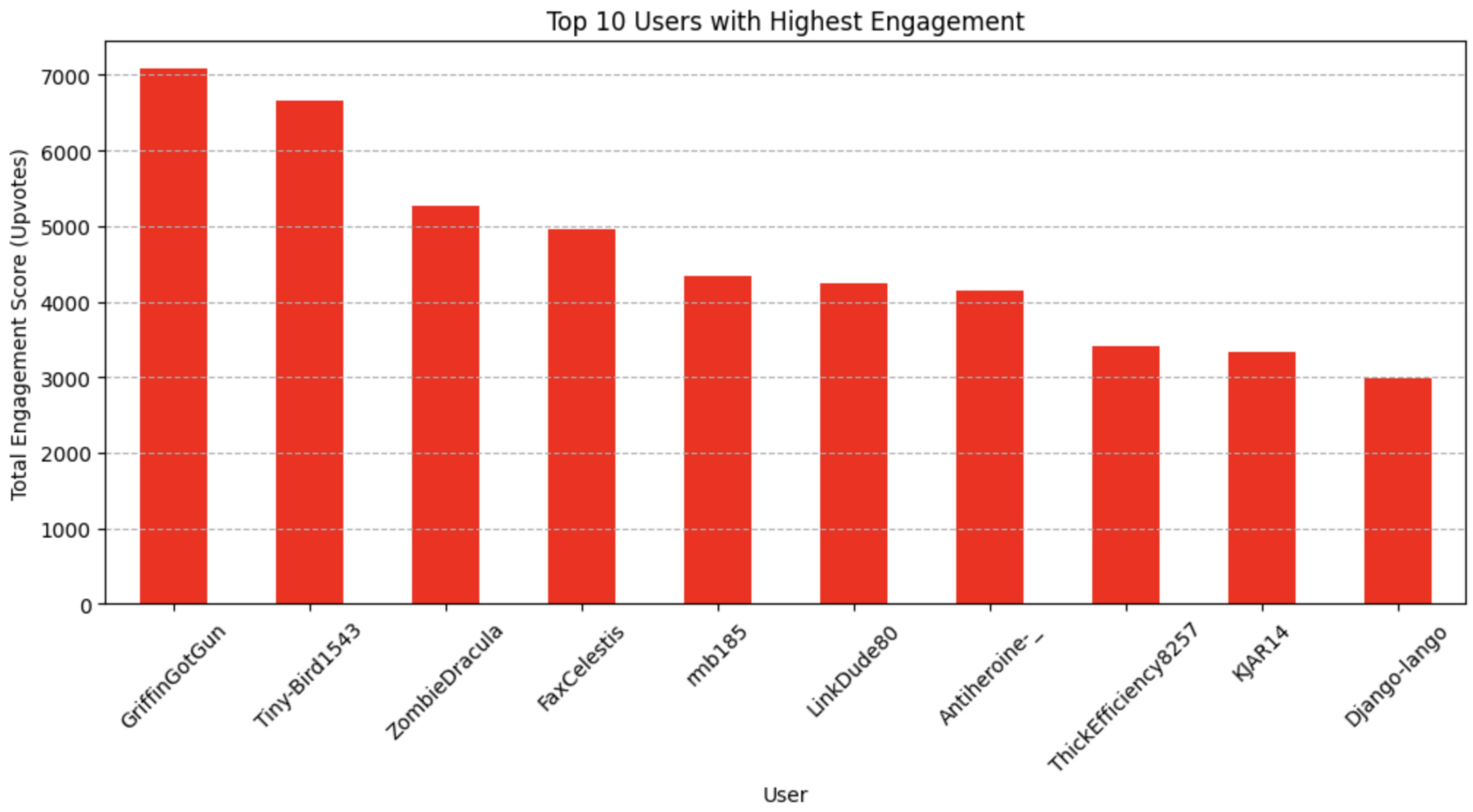
Daily Post Frequency Over Time by Disease Category



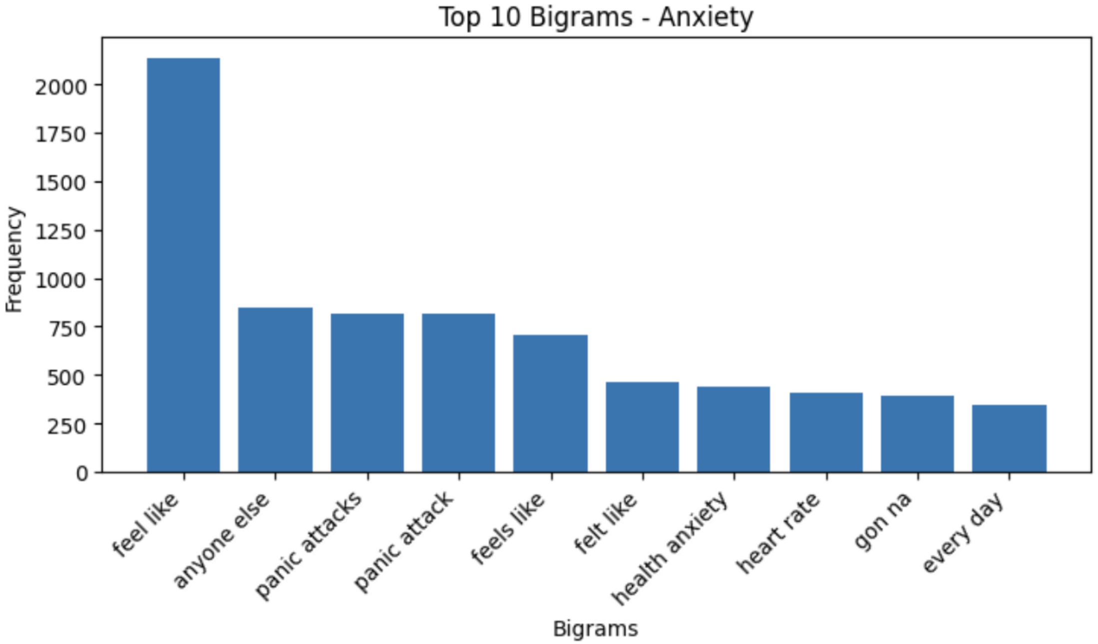
Top 10 Active Users

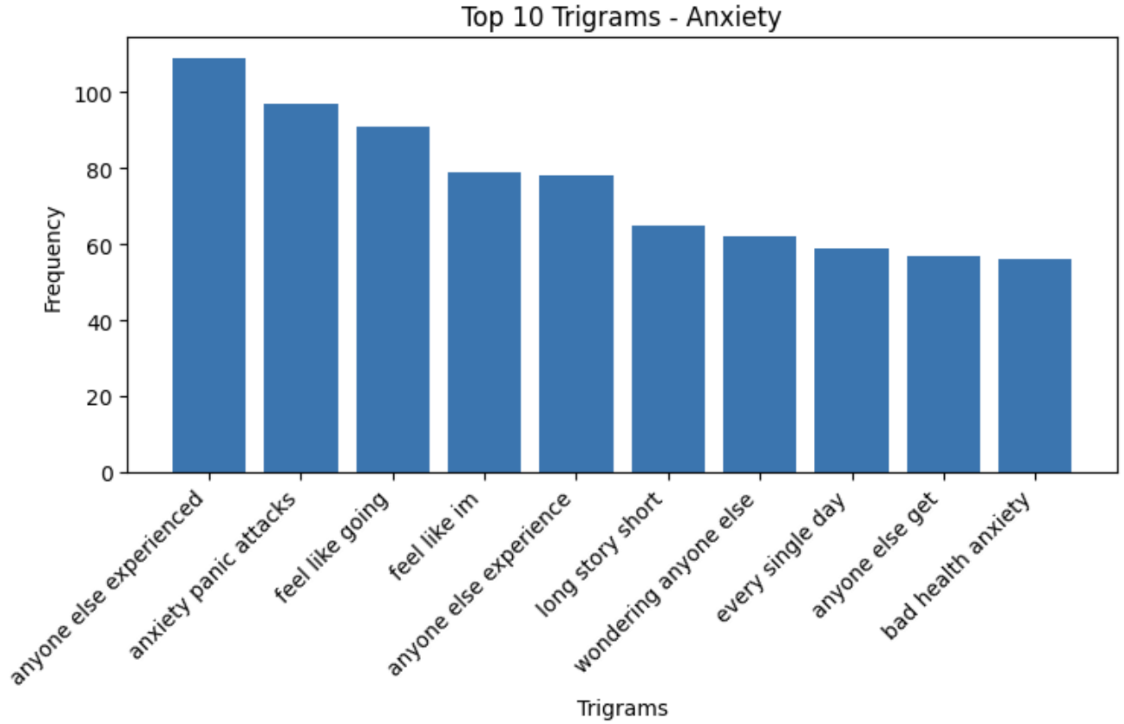


Top 10 Users with Highest Engagement

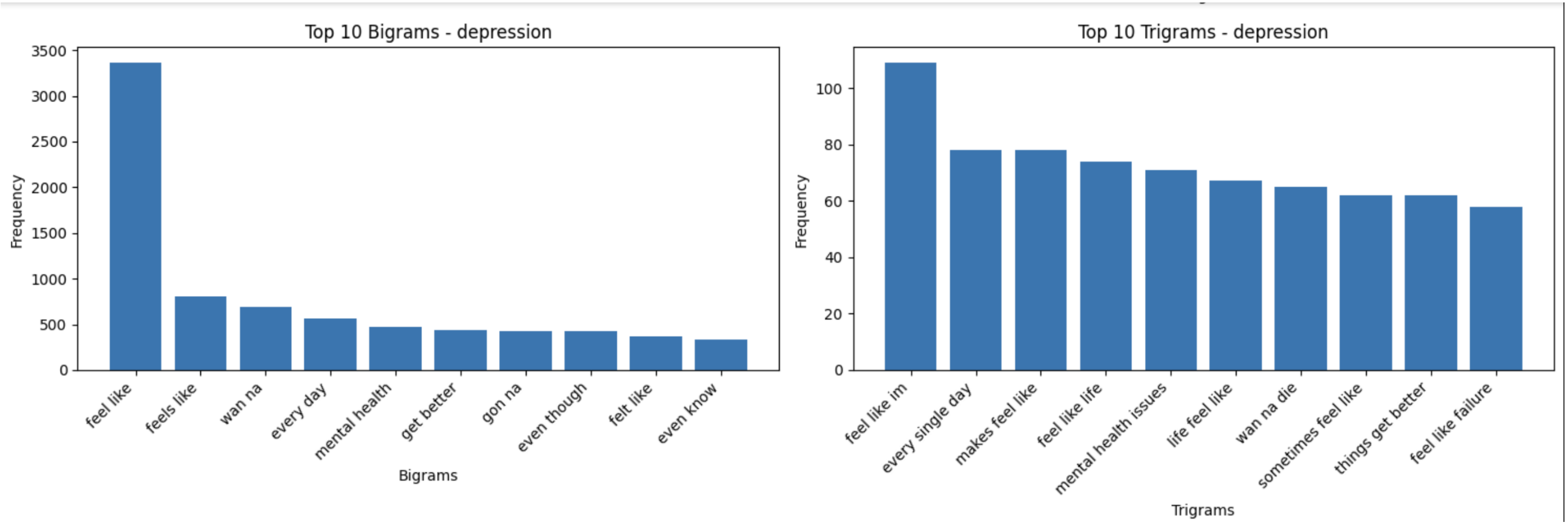


Top 10 Ngrams for Anxiety

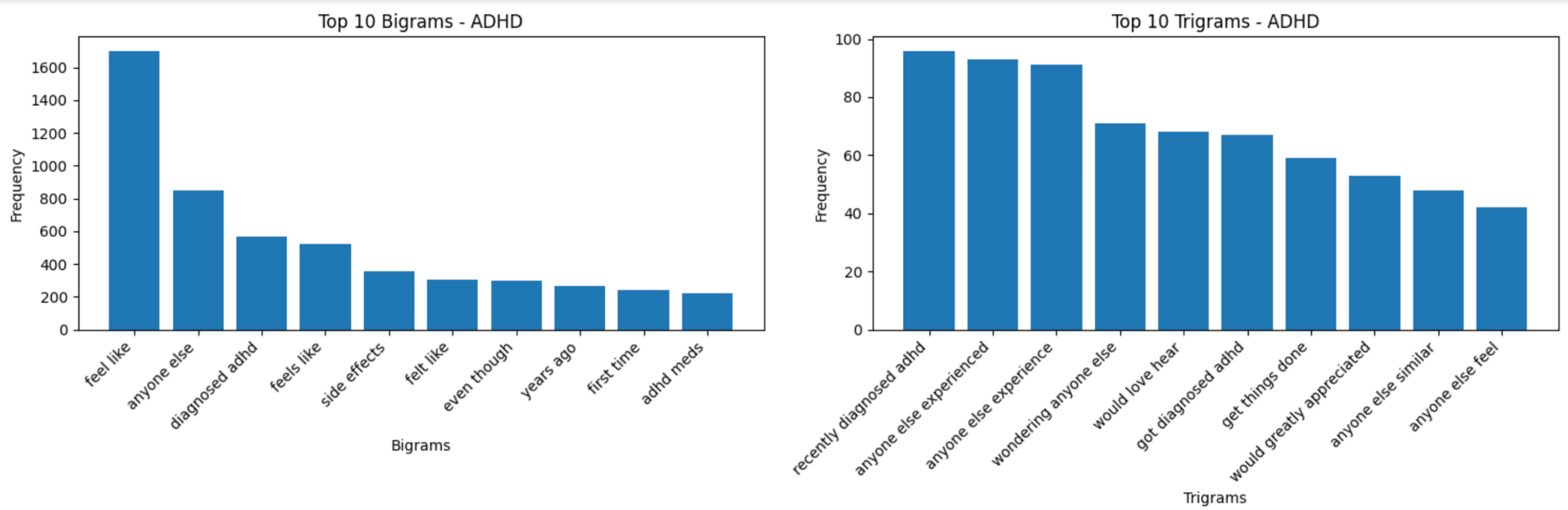




Top 10 Ngrams for Depression



Top 10 Ngrams for ADHD



Top 10 Users with their Subreddits post

Feature Engineering

Text Processing:

- **Tokenization:** Split text into individual words.
- **Stopword Removal:** Removed common but uninformative words.
- **Lemmatization:** Reduced words to their base forms to simplify analysis.

New Features Created:

  - 'New_Token_Selftext': Cleaned and tokenized text.
  - 'Lemmatized_Text': Further processed for base word forms.
  - 'Merged_Lemma_Token': Combined clean tokens for comprehensive text analysis.

- **Medication Analysis:**
  - Identified and counted mentions of anxiety, depression, and ADHD medications.
  - Generated word clouds to visualize commonly discussed medications.
- **Engagement Metrics:**
  - Analyzed post scores and user activity levels to understand engagement trends.
  - Examined top-performing posts based on engagement scores.

Severity Analysis Model

Text Processing:

- The goal is to predict mental health disorder category (Anxiety, Depression, ADHD) and severity from user selftext.

Approach:

- **Keyword based Labelling** : Assign label based on disorder specific keyword presence
- **Random Forest Classifier** – to predict disorder category
- **Random Forest Regressor** – to predict severity score

Assigning Predicted Labels

- **Keyword-Based Labeling**
  - Each disorder has a keyword set (e.g., "hopeless" → Depression, "racing thoughts" → Anxiety).
  - If multiple keywords appear, the highest-scoring disorder is chosen
  - Subreddit Influence – if no strong keyword signal then default to the subreddit label
- **Machine Learning Enhancement**
  - TF-IDF vectorization (Bigrams and Trigrams)
  - Random Forest Classifier trained simulated labels
  - Prediction based on learned text patterns beyond simple keywords

Assigning Severity Scores

- **Base Score Calculation**
  - Minimum score: 4 (Depression, ADHD) or 5 (Anxiety)
  - Keyword frequency increases severity score.
  - **Modifiers**:
    - Intensifiers: Words like "*severe*", "*every day*", "*extreme*" → Increase Score
    - Qualifiers: Words like "*sometimes*", "*mild*" → Decrease Score
    - Medication Mention: Presence of medication names (e.g., Prozac, Adderall) raises severity.
- **Final Steps**
  - Random Forest Regressor refines severity prediction
  - Ensures scores fall within valid disorder-specific severity ranges

How Rule-Based Scoring Works?

Assigns score (1-10) to train the ML model.

Selftext Example : "I have severe chest pain and can't stop worrying"

1. Disorder Range: Anxiety → 5-10

2. Base Score: Starts at 1

   - Keywords: "*chest pain*", "*can't stop worrying*" (2 found) → jumps to 5 (min for Anxiety).

3. Adjustments:

   - Intensifier: "severe" → +1 (5 → 6).
   - No qualifiers (e.g., "mild") → no subtraction.
   - No medications (e.g., "xanax") → no further change.

4. Constraint: 6 fits within 5-10 → Final Score = 6.

- **Text → Keywords (2) → Base = 5 → "Severe" (+1) → Score = 6**

```
df_new[['Selftext', 'Subreddit','Training_Disorder', 'Predicted_Disorder'
,'Simulated_Severity', 'Predicted_Severity']].head(5)
```

	Selftext	Subreddit	Training_Disorder	Predicted_Disorder	Simulated_Severity	Predicted_Severity
0	Hi.\n\nLike many I suffer from bad health anxi...	Anxiety	Depression	Depression	4	5
1	I was prescribed propranolol for my anxiety. ...	Anxiety	Anxiety	Anxiety	6	6
2	Our house is 1970s and it’s very likely it has...	Anxiety	Anxiety	Anxiety	5	5
3	Is anyone waking up feeling weak or drained? I...	Anxiety	Anxiety	Anxiety	5	5
4	I am going nuts.\nMy boyfriend and I broke up ...	Anxiety	Depression	Depression	4	4

Output

- Selftext: Contains the full text of user posts (e.g., personal experiences or symptoms), used as input for disorder and severity analysis.
- Subreddit: Indicates the source subreddit (e.g., "Anxiety"), reflecting the original context or community of the post.
- Training_Disorder: The disorder label used to train the model (e.g., "Depression"), based on subreddit or other initial classification.
- Predicted_Disorder: The disorder predicted by the machine learning model (e.g., "Depression"), derived from text analysis.
- Simulated_Severity: A rule-based severity score (1-10) generated by simulate_severity , serving as the training target for the model.
- Predicted_Severity: The final severity score (1-10) predicted by the Random Forest Regressor, adjusted to disorder-specific ranges.

Disadvantages of Assigning and Predicting Severity Score

- **Overfitting**: Random Forest may memorize training data (e.g., Simulated_Severity), inflating accuracy and limiting generalization.
- **Data Leakage**: Rule-based features (e.g., keywords in TF-IDF) may overlap with targets, artificially boosting performance.
- **Imbalanced Dataset**: High accuracy might reflect bias toward common severity ranges, missing rare cases.
- **Metric Limitation**: Accuracy hides error magnitude (e.g., 4 vs. 5 vs. 9), misrepresenting prediction quality.

Challenges, Limitations & Next Steps

Challenges & Limitations:

Reddit API does not provide access to historical data due to recent policy changes

PRAW API is limited to retrieving only the past 10–15 days of data

Mitigation Strategies:

Optimizing data fetch cycles to capture the freshest data available

Monitoring API changes to adapt the data acquisition process

Next Steps:

Focus on creating models for training and predicting the severity and disease based on the text

Collaboration With Experts

To ensure the accuracy and reliability of our mental health predictions, we are collaborating with industry experts who bring valuable domain knowledge and insights:

Ashleigh Zoerb – Licensed Mental Health Therapist (10+ years of experience)

- Provides expertise in assessing mental health severity.
- Helps validate the severity classification algorithm based on real-world mental health cases.

Nelca Ece Yilmaz – Masters in Psychology

- Provides expertise in assessing mental health severity.
- Helps validate the severity classification algorithm based on real-world mental health cases.

Their guidance ensures that our model aligns with professional mental health assessments, enhancing the credibility and impact of our research.

Key Lessons Learnt

- Data Extraction through API
- Data Acquisition
- Data Preprocessing
- Exploratory Data Analysis
- How to create a data pipeline
- How to deal with unstructured data

THANK YOU!!